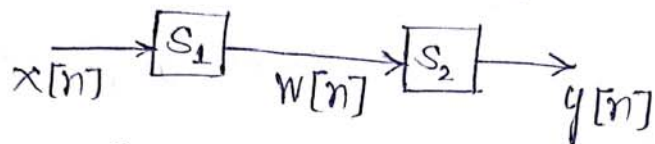


Sol 1(a):

Given that - $w[n] = \frac{1}{2}w[n-1] + x[n]$ — (1)

$y[n] = \alpha y[n-1] + \beta w[n]$ — (2)

$y[n] = -\frac{1}{8}y[n-2] + \frac{3}{4}y[n-1] + x[n]$ — (3)



S_1, S_2 : Causal LTI system

$\therefore y[n] = \alpha y[n-1] + \beta w[n]$

From this we may write,

$w[n] = \frac{1}{\beta} y[n] - \frac{\alpha}{\beta} y[n-1]$ — (4)

and

$w[n-1] = \frac{1}{\beta} y[n-1] - \frac{\alpha}{\beta} y[n-2]$ — (5)

Multiplying both side of eqⁿ(5) with $(\frac{1}{2})$, we get -

$\frac{1}{2}w[n-1] = \frac{1}{2\beta} y[n-1] - \frac{\alpha}{2\beta} y[n-2]$ — (6)

Subtracting eqⁿ(4) & eqⁿ(6), we get -

$w[n] - \frac{1}{2}w[n-1] = \frac{1}{\beta} y[n] - \frac{\alpha}{\beta} y[n-1] - \frac{1}{2\beta} y[n-1] + \frac{\alpha}{2\beta} y[n-2]$ — (7)

By eqⁿ(1) & eqⁿ(7), we get -

$x[n] = \frac{1}{\beta} y[n] - \frac{\alpha}{\beta} y[n-1] - \frac{1}{2\beta} y[n-1] + \frac{\alpha}{2\beta} y[n-2]$

$y[n] = (\alpha + \frac{1}{2}) y[n-1] - (\frac{\alpha}{2}) y[n-2] + \beta x[n]$ — (8)

Comparing eqⁿ(3) and eqⁿ(8), we get —

$$\left(\alpha + \frac{1}{2}\right) = \left(\frac{3}{4}\right) \text{ and } \beta = 1$$

$$\alpha = \left(\frac{1}{4}\right) \text{ and } \beta = 1 \rightarrow (2 \times 2 \text{ Marks})$$

Sol 1(b):

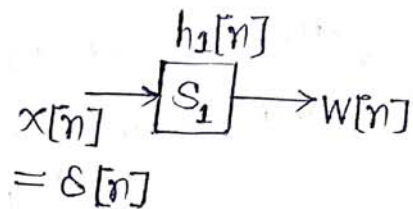
Let impulse response of system S_1 & S_2 are $h_1[n]$ & $h_2[n]$ respectively.

By eqⁿ(1), $W[n] = \frac{1}{2} W[n-1] + x[n]$

$$h_1[n] - \frac{1}{2} h_1[n-1] = \delta[n]$$

$$\therefore h_1[n] = \left(\frac{1}{2}\right)^n u[n] \quad \text{--- (9)}$$

$\rightarrow (1 \text{ Marks})$



$$\begin{aligned} W[n] &= x[n] * h_1[n] \\ &= \delta[n] * h_1[n] \\ &= h_1[n] \end{aligned}$$

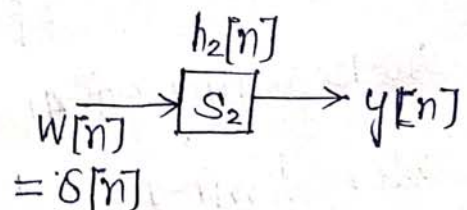
By eqⁿ(2), $Y[n] = \alpha Y[n-1] + \beta W[n]$

$$Y[n] = \frac{1}{4} Y[n-1] + W[n]$$

$$h_2[n] - \frac{1}{4} h_2[n-1] = \delta[n]$$

$$\therefore h_2[n] = \left(\frac{1}{4}\right)^n u[n] \quad \text{--- (10)}$$

$\rightarrow (1 \text{ Marks})$



$$\begin{aligned} Y[n] &= W[n] * h_2[n] \\ &= \delta[n] * h_2[n] \\ &= h_2[n] \end{aligned}$$

$\therefore$ Impulse response of cascaded systems,

$$h[n] = h_1[n] * h_2[n]$$

$$= \sum_{k=-\infty}^{\infty} h_1[k] \cdot h_2[n-k]$$

$$\begin{aligned}
 \therefore h[n] &= \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k} u[n-k] \\
 &= \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k} = \left(\frac{1}{4}\right)^n \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{-k} \\
 &= \left(\frac{1}{4}\right)^n \sum_{k=0}^n \left(\frac{1}{2}\right)^{-k} = \left(\frac{1}{4}\right)^n \cdot \frac{1[2^{n+1} - 1]}{[2 - 1]} \\
 &= \left(\frac{1}{2}\right)^{2n} [2^{n+1} - 1] \\
 &= \left[2\left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n\right] u[n] \quad \rightarrow (2 \text{ Marks})
 \end{aligned}$$

Sol(2)÷ Given impulse response of an LTI system,

$$h(t) = e^{-6|t|}$$

$$= \begin{cases} e^{-6t}, & t > 0 \\ e^{-6(-t)}, & t < 0 \end{cases}$$

$$= e^{-6t} \cdot u(t) + e^{6t} \cdot u(-t) \quad \text{---(1)}$$

Step(I)÷ Condition for LTI system to be causal,
 $h(t) = 0, \quad t < 0$

By eqⁿ(1), $h(t) \neq 0$ for $t < 0$, so the given LTI system is Non-causal.

→ (2 Mark)

Step(II)÷ Condition for LTI system to be stable, impulse response of the system should be absolutely integrable.

$$\text{i.e. } \int_{-\infty}^{\infty} |h(t)| dt < \infty$$

$$\text{By eqⁿ(1), } h(t) = e^{-6t} u(t) + e^{6t} u(-t)$$

$$\int_{-\infty}^{\infty} |h(t)| dt = \int_{-\infty}^{\infty} |e^{-6t} u(t) + e^{6t} u(-t)| dt$$

$$= \int_{-\infty}^0 e^{6t} dt + \int_0^{\infty} e^{-6t} dt = \left(\frac{e^{6t}}{6} \right)_{-\infty}^0 + \left(\frac{e^{-6t}}{-6} \right)_0^{\infty}$$
$$= \left(\frac{1-0}{6} \right) + \left(\frac{0-1}{-6} \right) = \left(\frac{1}{3} \right)$$

So, the given LTI system is stable.

→ (2 Mark)

Sol 3(I)÷ Given that- $x(t) = 3 \cos(4t + \frac{\pi}{3})$ — (1)

For Periodic signal,

$$x(t+T) = x(t) \quad \forall t$$

where, $T = \text{FTP}$ (smallest positive)

$$\begin{aligned} x(t+T) &= 3 \cos \left\{ 4(t+T) + \frac{\pi}{3} \right\} \\ &= 3 \cos \left\{ 4t + \frac{\pi}{3} + 4T \right\} \end{aligned}$$

if $T = \frac{\pi}{2}$

$$\begin{aligned} \text{then— } x(t+T) &= 3 \cos \left\{ 4t + \frac{\pi}{3} + 4 \times \frac{\pi}{2} \right\} \\ &= 3 \cos \left\{ 4t + \frac{\pi}{3} + 2\pi \right\} \\ &= 3 \cos \left(4t + \frac{\pi}{3} \right) \\ &= x(t) \end{aligned}$$

→ (1.5 Mark)

Hence, $x(t)$ is periodic signal with FTP, $T = \frac{\pi}{2}$

Sol 3(II)÷ Given that- $x[n] = \cos(\frac{\pi}{2}n) \cos(\frac{\pi}{4}n)$ → (1 Mark)

$$\begin{aligned} &= \frac{1}{2} \left\{ \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}n\right) + \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}n\right) \right\} \\ &= \frac{1}{2} \left\{ \cos\left(\frac{3\pi}{4}n\right) + \cos\left(\frac{\pi}{4}n\right) \right\} \\ &= \frac{1}{2} \cos\left(\frac{3\pi}{4}n\right) + \frac{1}{2} \cos\left(\frac{\pi}{4}n\right) \\ &= x_1[n] + x_2[n] \quad \text{— (2)} \end{aligned}$$

For Periodic signal,

$$x[n+N] = x[n] \quad \forall n$$

where, $N = \text{FTP}$ (smallest integer)

(0.5 Mark)

By eqⁿ(2), $x_1[n] = \frac{1}{2} \cos\left(\frac{3\pi}{4}n\right)$

$$\begin{aligned}x_1[n+N_1] &= \frac{1}{2} \cos\left\{\frac{3\pi}{4}(n+N_1)\right\} \\&= \frac{1}{2} \cos\left\{\frac{3\pi}{4}n + \frac{3\pi}{4}N_1\right\}\end{aligned}$$

If $N_1 = 8$ then — $x_1[n+N_1] = x_1[n]$

Hence, $x_1[n]$ is periodic signal with FTP, $N_1 = 8$

— (3)
→ (1 Mark)

By eqⁿ(2), $x_2[n] = \frac{1}{2} \cos\left(\frac{\pi}{4}n\right)$

$$\begin{aligned}x_2[n+N_2] &= \frac{1}{2} \cos\left\{\frac{\pi}{4}(n+N_2)\right\} \\&= \frac{1}{2} \cos\left\{\frac{\pi}{4}n + \frac{\pi}{4}N_2\right\}\end{aligned}$$

If $N_2 = 8$ then — $x_2[n+N_2] = x_2[n]$

Hence, $x_2[n]$ is periodic signal with FTP, $N_2 = 8$

— (4)
→ (1 Mark)

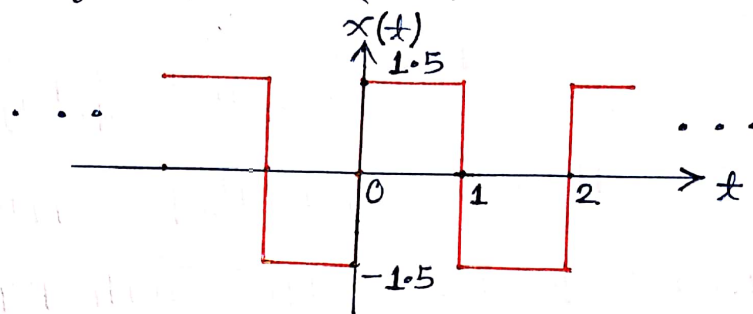
By eqⁿ(2), (3) & (4), we get —

$x[n]$ is periodic signal with FTP, $N = \text{LCM}(N_1, N_2) = 8$

→ (0.5 Mark)

Sol(4)÷ Given that -

$$x(t) = \begin{cases} 1.5, & 0 \leq t < 1 \\ -1.5, & 1 \leq t < 2 \end{cases}$$



(a) Fundamental Time period of signal $x(t)$, $T_0 = 2$

Fundamental freq., $\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2} = \pi \text{ rad/sec}$
→ (1 Mark)

(b) As we know - Fourier series coefficients a_k for periodic signal $x(t)$,

$$a_k = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^{-jk\omega_0 t} dt$$

Case(I)÷ a_k for $k=0$, $a_0 = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^0 \cdot dt$
 $= \frac{1}{2} \int_0^2 x(t) dt = 0$ → (1 Mark)

Case(II)÷ a_k for $k \neq 0$, $a_k = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^{-jk\omega_0 t} dt$
 $= \frac{1}{2} \int_0^2 x(t) \cdot e^{-jk\pi t} dt$
 $= \frac{1}{2} \left\{ \int_0^1 1.5 e^{-jk\pi t} dt + \int_1^2 (-1.5) e^{-jk\pi t} dt \right\}$

$$\begin{aligned}
a_k &= \frac{1}{2} \left\{ 1.5 \left(\frac{e^{-j\pi k t}}{-j\pi k} \right) \Big|_0^1 - 1.5 \left(\frac{e^{-j\pi k t}}{-j\pi k} \right) \Big|_1^2 \right\} \\
&= \frac{1}{2} \left\{ \left(\frac{1.5}{-j\pi k} \right) (e^{-j\pi k} - 1) - \left(\frac{1.5}{-j\pi k} \right) (e^{-j\pi k \cdot 2} - e^{-j\pi k}) \right\} \\
&= \frac{1}{2} \left(\frac{1.5}{-j\pi k} \right) \left\{ e^{-j\pi k} - 1 - e^{-j2\pi k} + e^{-j\pi k} \right\} \\
&= \frac{1}{2} \left(\frac{1.5}{-j\pi k} \right) \left\{ 2e^{-j\pi k} - e^{-j2\pi k} - 1 \right\} \\
&= \frac{1}{2} \left(\frac{1.5}{-j\pi k} \right) (2e^{-j\pi k} - 1 - 1) = \frac{1}{2} \left(\frac{1.5}{-j\pi k} \right) (2e^{-j\pi k} - 2) \\
&= \left(\frac{1.5}{j\pi k} \right) (1 - e^{-j\pi k}) \\
&= \left(\frac{3}{k\pi} \right) \left(\frac{e^{-j\frac{\pi k}{2}} \cdot e^{j\frac{\pi k}{2}} - e^{-j\frac{\pi k}{2}} \cdot e^{-j\frac{\pi k}{2}}}{2j} \right) \\
&= \left(\frac{3}{k\pi} \right) e^{-j\frac{\pi k}{2}} \cdot \sin\left(\frac{\pi k}{2}\right) \quad \rightarrow (4 \text{ Mark})
\end{aligned}$$